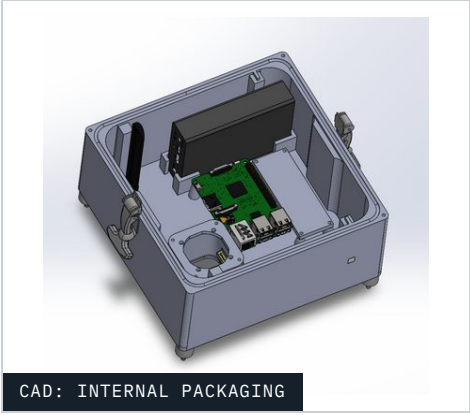
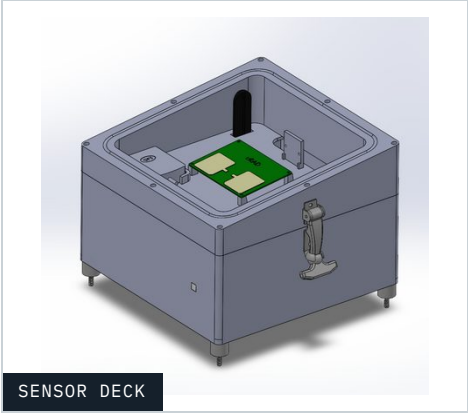
01 NORTHERN HAIL PROJECT ENCLOSURE

Field-deployable enclosure for radar-based hail sensing and environmental data

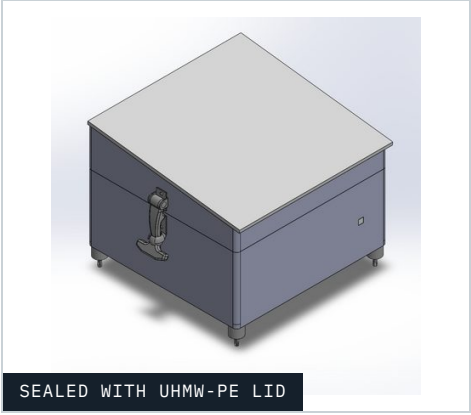
CONTEXT	MY ROLE	TOOLS
Arcadian Research & Contracting, for Western's Northern Hail Project	Co-founder and research engineer	SOLIDWORKS, FEA, GD&T, PETG/ASA printing, UHMW-PE machining, O-ring seal design



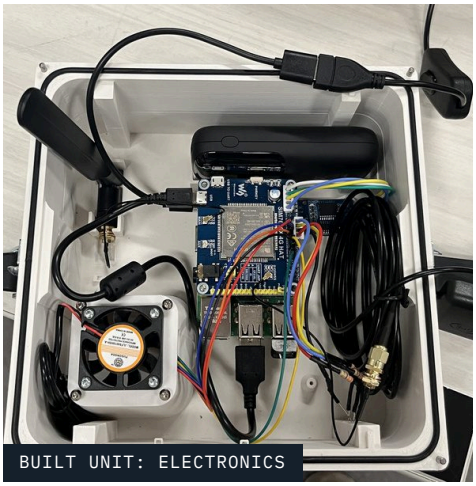
CAD: INTERNAL PACKAGING



SENSOR DECK



SEALED WITH UHMW-PE LID



BUILT UNIT: ELECTRONICS



BUILT UNIT: RADAR DECK



PRODUCTION BATCH

OVERVIEW

A sealed enclosure for radar instrumentation and environmental sensors, built to survive hail, rain and wind in the field.

WHAT I DID

- Designed a two-part printed enclosure with an O-ring groove and latch geometry, validated to IP67-equivalent waterproofing (30-minute submersion)
- Applied GD&T to sealing and dowel-pin interfaces, holding ±0.1 mm on critical fits for consistent gasket compression
- Used FEA to identify and resolve structural failure points in early prototypes while maintaining the seal
- Optimized the UHMW-PE lid for radar transparency, machining efficiency and water runoff
- Integrated mounts for the Raspberry Pi, modem and sensors with clean cable routing and assembly access
- Built machining jigs for sheet cutting and drilling to keep production units consistent

RESULTS

IP67-equiv.
30-minute submersion test

4 in.
Simulated hail impact survived

Tornado
A field unit was caught in one and kept functioning

02 TENDON-DRIVEN ROBOTIC HANDS

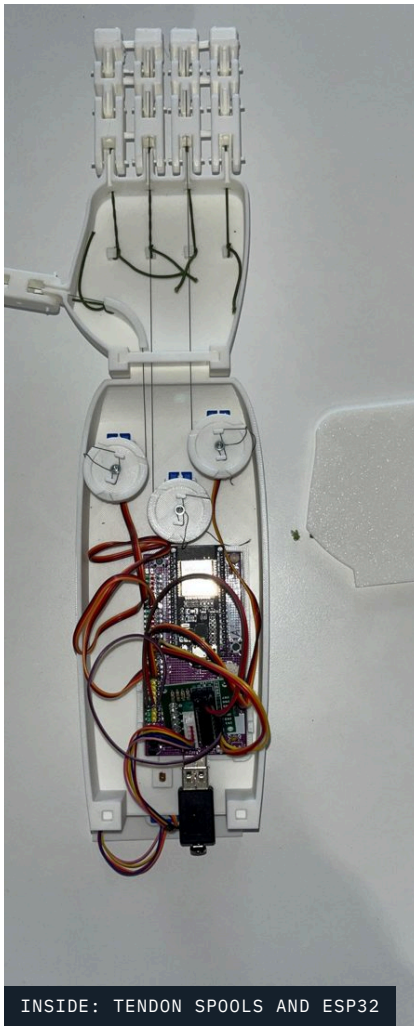
Two 3D-printed hands that play Rock-Paper-Scissors-Minus-One against each other

CONTEXT	MY ROLE	TOOLS
Sensors and Actuators course, Western (team project)	Mechanical design, 3D printing and most of the electrical work	SOLIDWORKS, 3D printing, servos, ESP32, BLE


ROCK


PAPER


SCISSORS


INSIDE: TENDON SPOOLS AND ESP32

OVERVIEW

Rock-Paper-Scissors-Minus-One is a two-stage variant of the game solved with a Nash equilibrium strategy. Each hand runs on an ESP32 that talks over BLE to a Python decision engine on a host computer.

WHAT I DID

- Led the mechanical design of two articulated hands, printed and assembled in-house
- Grouped the tendons so one servo drives each gesture rather than each finger, cutting the actuator count
- Added a passive elastic return so the fingers reopen without extra actuators
- Handled most of the electrical work, including the servo and ESP32 wiring

RESULTS

93.3%
Gesture accuracy across physical trials

1 servo
Per gesture, not per finger

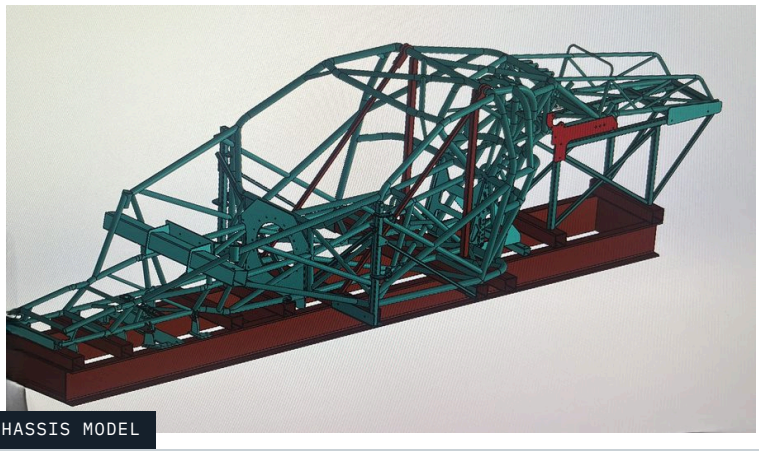
<20 s
Every move made inside the game's time window

colebook9@gmail.com · 226-938-1472 · linkedin.com/in/cole-book

03 DRAG CAR CHASSIS

Welded-tube chassis designed for manufacturability, stiffness and composite body integration

CONTEXT	MY ROLE	TOOLS
Cynergy Composites, mechanical design co-op	Chassis design and first-build support	SOLIDWORKS weldments, GD&T, DFM, CNC tube notching prep



CHASSIS MODEL



FABRICATION



FIRST BUILD ON THE JIG

OVERVIEW

A modular NHRA drag car chassis modeled as a SOLIDWORKS weldment and set up for CNC tube notching, fixture assembly and composite bodywork.

WHAT I DID

- Modeled the 200+ tube chassis in the weldment environment with automated cut lists and part numbering
- Reduced total part count by 10% while meeting NHRA structural requirements
- Applied GD&T to mounting points and suspension geometry to hold alignment during welding and jig setup
- Designed CNC tube templates and fixture points for consistent production
- Worked with the composite body and rear-end teams on mounting interfaces and packaging
- Troubleshot fit-up and modified weld sequences on the shop floor during the first build

RESULTS

200+
Tubular members in one weldment model

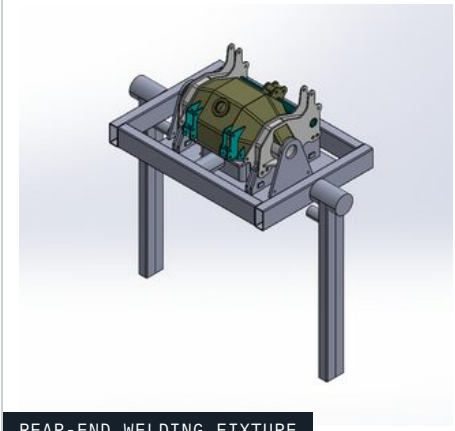
–10%
Total part count

NHRA
Built to drag racing safety standards

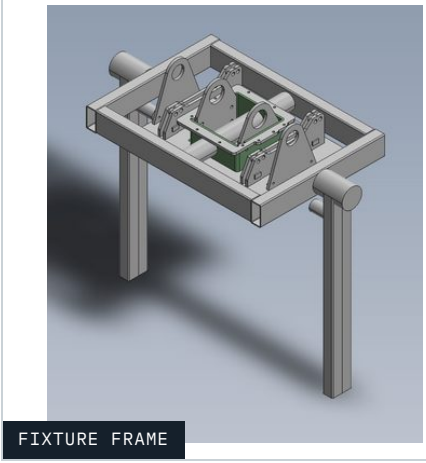
04 TOOLING, MOLDS & FUEL TANK

Welding fixtures, composite mold tooling and a custom fuel tank for a drag car program

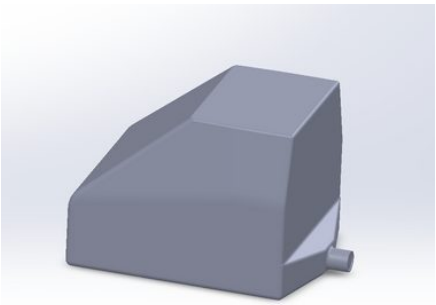
CONTEXT	MY ROLE	TOOLS
Cynergy Composites, mechanical design co-op	Fixture, tooling and sheet metal design; reverse engineering	SOLIDWORKS surfacing, GD&T, tolerance analysis, sheet metal, 3D scanning



REAR-END WELDING FIXTURE



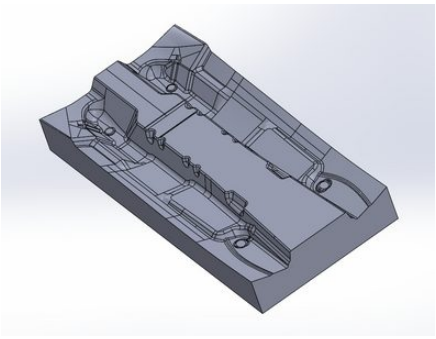
FIXTURE FRAME



ALUMINUM FUEL TANK



DOOR PANEL SURFACES



MOLD TOOL

OVERVIEW

Tooling and components for a high-performance drag car program, focused on repeatable geometry, manufacturability and accurate surfaces for composite parts.

WHAT I DID

- Designed and fabricated a rear-end welding fixture that holds tube and bracket alignment through tack and final welds
- Developed Ford GT door panel mold tooling with surfacing, parting-line control and draft management, using GD&T and tolerance analysis to improve dimensional accuracy
- Designed a custom aluminum fuel tank with internal baffles, a welded filler neck and mounting tabs
- Reverse engineered a stock car body and velocity stacks with 3D scanning and scan-to-CAD surfacing

RESULTS

±0.5 mm
Scan-to-CAD accuracy on reverse engineered parts

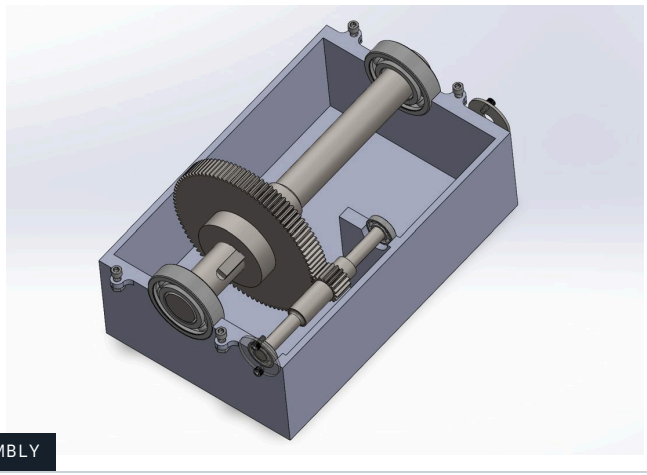
Ford GT
Door panel tooling, more accurate than the previous generation

1st build
Fixture used on the first custom chassis

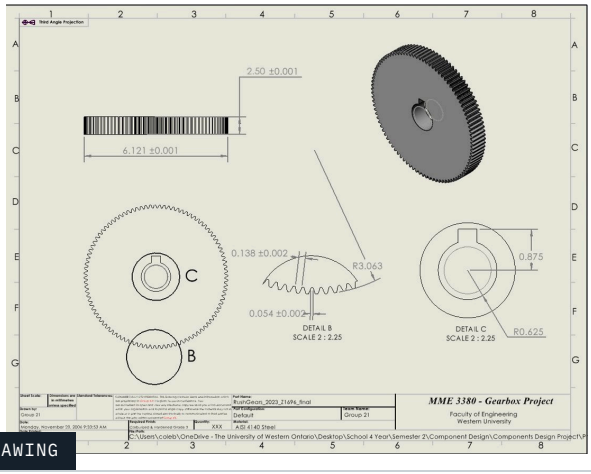
05 REHABILITATION TREADMILL GEARBOX

6:1 single-stage spur gear reducer for a low-speed treadmill for injured and elderly users

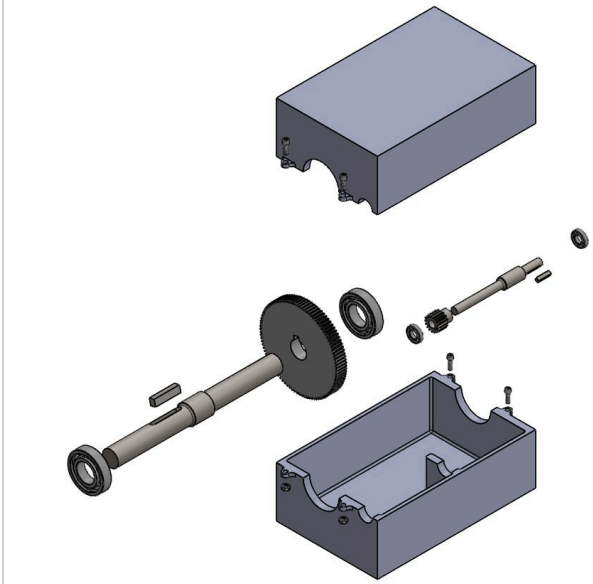
CONTEXT	MY ROLE	TOOLS
Mechanical Component Design course, Western (team project)	CAD, drawings, bearing selection and shaft deflection analysis	SOLIDWORKS, shaft fatigue and deflection analysis, bearing life calculations



ASSEMBLY



GEAR DRAWING



EXPLODED VIEW

OVERVIEW

The gearbox had to deliver a 5 mph belt speed under a 400 lbf user load with a 20,000-mile service life inside a fixed housing envelope.

WHAT I DID

- Produced the SOLIDWORKS CAD, assembly and part drawings
- Found the first design failed: the 0.5 in. pinion shaft spanning the full 12.25 in. housing exceeded deflection and slope limits, with a fatigue factor of safety of 0.41
- Added an internal support wall to cut the bearing span to 6.5 in. and upgraded to quenched and tempered 4140 steel, bringing both shafts within limits
- Sized deep-groove ball bearings for over 400 million pinion revolutions
- Documented that the pinion still missed the 2.5 target, traced it to the single-stage ratio and housing size, and recommended a larger housing or two-stage reduction

RESULTS

0.41 → 1.72
Pinion fatigue factor of safety

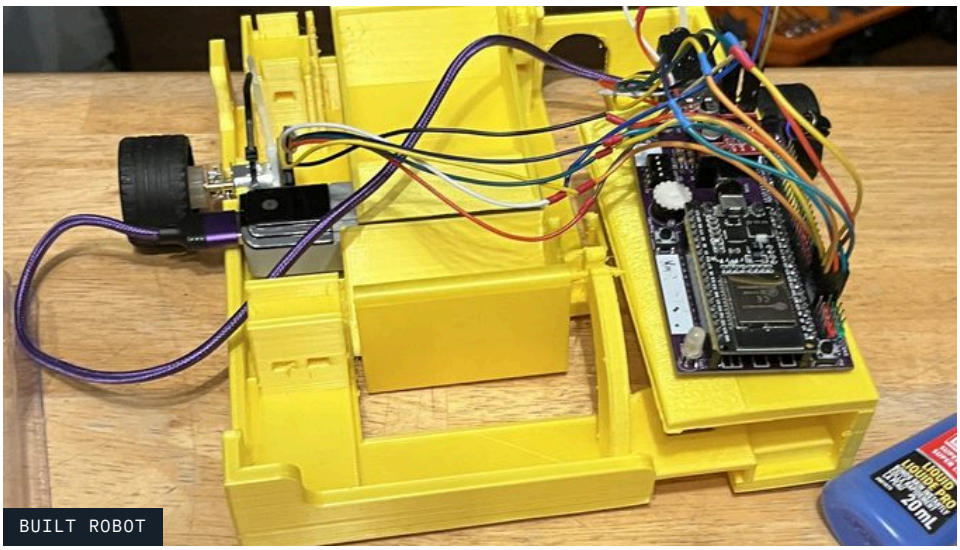
12.25 → 6.5 in.
Pinion bearing span

400M+
Pinion revolutions of bearing life

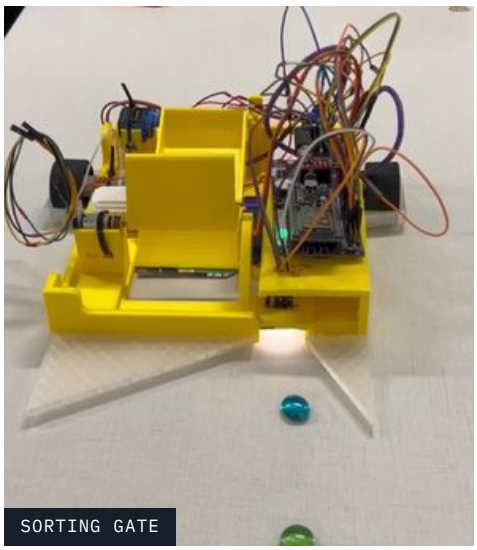
06 AUTONOMOUS SORTING ROBOT

ESP32 mobile robot that collects, color-sorts and returns marbles in a 120-second run

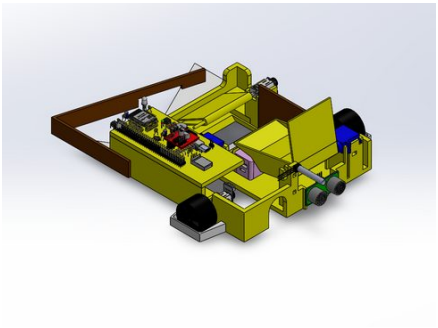
CONTEXT	MY ROLE	TOOLS
Introduction to Mechatronic Design course, Western	Chassis design and C++ control code	SOLIDWORKS, 3D printing, ESP32, C++, PWM, encoders, ultrasonic, IR and color sensing



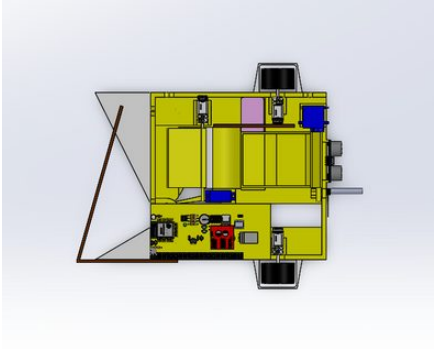
BUILT ROBOT



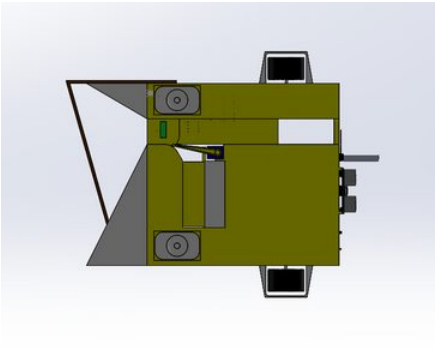
SORTING GATE



CHASSIS CAD



TOP LAYOUT



UNDERSIDE

OVERVIEW

The robot navigates an arena, collects marbles, identifies them by color, sorts them onboard and returns the selected ones to a base station.

WHAT I DID

- Designed the 3D-printed chassis and component mounts, laying them out to cut print time and material while keeping the motor mounts rigid
- Wrote the C++ for differential steering with PWM motor control and wheel encoder feedback
- Added ultrasonic obstacle sensing and IR beacon homing to find the base station
- Used a color sensor to drive a servo gate that accepts or rejects each marble
- Used non-blocking millis() timing so sensing and motor control run concurrently on one microcontroller

RESULTS

95%
Sorting accuracy

120 s
Full collect, sort and return cycle

1 MCU
Sensing and actuation running concurrently